Session 7: CPSA formalism, and Beyond CPSA

Slides prepared by UMBC Protocol Analysis Lab

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- Proofs *about* theorem provers are extremely complicated
  - Important if you are interested in proving things *about* CPSA, rather than *with* CPSA.

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- Scenario + Assumptions + Set of actions possible = analysis
- "Enrichment" refers to finding plausible actions to explain the scenario
- Once the scenario is explained, then a skeleton is called *realized*

Post Quantum Cryptography

What the future of crypto looks like, but when?

Beyond CPSA

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Post-quantum attacks are not abstracted from the primitive, so CPSA cannot consider them.

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Grover's algorithm has, so far, not been shown to be feasible.

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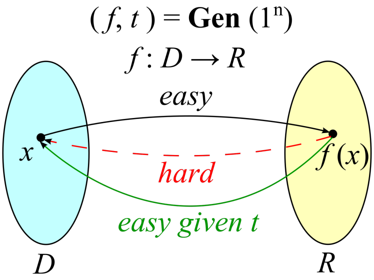
- CRYSTALS-Kyber chosen as the round winner for Key Encapsulation (KEM).
- Lattice problem as trapdoor, average case NP-hard



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- Symmetric key schemes can be vulnerable to other attacks

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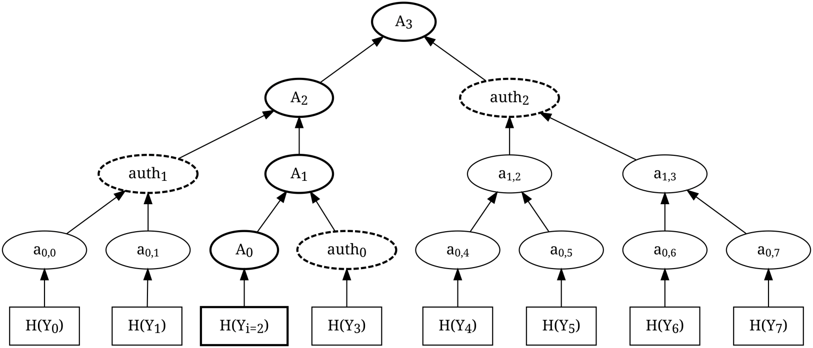

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